

JOHOR PORT BERHAD

Image: mmcports.com.my/johor-port/

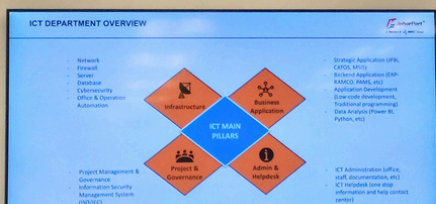
A WARM WELCOME

8 JANUARY 2026 — As part of the Technology Information System subject (SECP 1513), an industrial visit to Johor Port Berhad was organized to expose students to the practical application of information systems within a large-scale industrial environment. The visit aimed to enhance students' understanding of how technology and integrated information systems support port operations, logistics management, and decision-making processes.

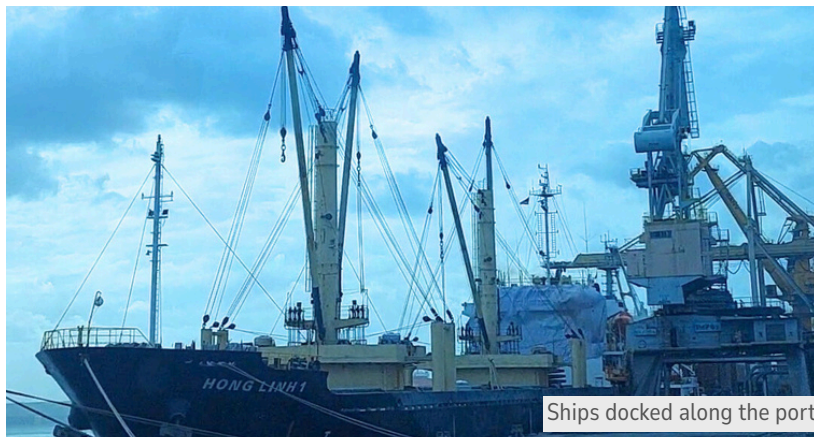
A TALK BY THE ICT DEPARTMENT

DEPARTMENT OVERVIEW

The ICT Department at Johor Port Berhad is responsible for managing and maintaining all information technology systems that support daily port operations. This includes network infrastructure, system security, data management, and operational software used for logistics, vessel tracking, and terminal management. The department ensures smooth communication between different departments and improves efficiency, safety and decision-making through digital solutions.



The main pillars of ICT



THE MAIN PILLARS OF ICT BUSINESS APPLICATION

The Business Application manages business operations through strategic and backend applications such as JPBI and PAMS. It also involves managing the application development, as well as data analysis like using Power BI and Python.

INFRASTRUCTURE

The Infrastructure pillar focuses on ICT components such as network, firewall, server, database and cybersecurity. Office & Operation Automation are also supported to ensure a stable IT environment.

ADMIN & HELPDESK

The Admin & Helpdesk pillar assists in ICT administration including office operations, staff coordination and documentation besides operating the ICT Helpdesk that is a one stop information and help contact center for ICT-related issues.

PROJECT & GOVERNANCE

The Project & Governance pillar is responsible for managing the project management activities and ensuring proper governance. It also supports the information security management system such as ISO/IEC to ensure a better ICT work processes.



BUSINESS APPLICATION HIGHLIGHTS



Image: mmcports.com.my/johor-port/

Johor Port employees use several applications developed by the ICT Department which are all vital in ensuring that everything runs smoothly within the port.

JPBI E-COMMUNITY

JPBI e-Community is a business application used by Johor Port employees as a main portal to access its strategic application systems such as the systems for the Container Terminal and Marine Services (MSIS).

ERP RAMCO

ERP stands for 'enterprise resource planning'. This is a software system which includes all necessary tools and processes needed for the port to run successfully.

PAMS

The port access management system manages access, permits and security for vehicles and people entering the port area.

The development for these applications use low-code platforms, which use drag-and-drop tools with very minimal code. These platforms are useful for when an application is needed urgently and is good for staying ahead of competing ports.

We were also introduced to the concept of data analysis, which is an important process when running a port. It refers to the process of inspecting, cleaning, transforming and modelling data to discover useful information.



Data Analysis

INFRASTRUCTURE HIGHLIGHTS



The port's infrastructure is made up of many different branches— data centres, networks, cybersecurity, and many more.

METROCLUSTER SOLUTION

The objective of this solution is to "achieve continuous data availability for mission critical application". It is designed for zero unplanned downtime through measures such as protection network faults and non-disruptive software and storage upgrades. A broadband network is also installed throughout the port as 50 kilometers of fiber connection.

CYBERSECURITY AND PERSONAL DATA PROTECTION

This sector provides advisory roles so employees are aware of the the necessary measures that should be taken to ensure satisfactory data protection and cybersecurity. Vulnerability assessments and penetration testing is also regularly carried out.

REFLECTING BACK

This visit had opened our eyes to the value Computer Science holds within the industry as the backbone of all operations. It improves productivity and increases profits by margins. From providing application systems to real-time data analysis, the ICT department of Johor Port proves that technological innovation is the way towards a brighter future.

We learned that even the most seemingly unimportant topic in a syllabus has its surprisingly common usage in the real world. To add, it is in the nature of IT to always be changing and evolving with time. So, in order to be ahead in any industry, knowledge should not be neglected and instead be constantly pursued. Following this method would guarantee an individual to remain valuable in a very competitive environment.

This has motivated us to not stray during our studies and to put in all of our effort into honing our skills in preparation for the future. One day, we hope our skills bring benefit and change to the world, improving lives one by one.

